Effat College of Engineering · Computer Science

CS3081 Artificial Intelligence
Assignment: Uncertainty and Bayesian Networks

Instructor: Dr. Naila Marir **Total:** 50 marks (5 exercises × 10 marks)

Name: Busra Suleymanoglu Student ID: S24109510

Instructions

1. Show all your steps. If you write only the final number, you get at most half of the marks.
2. Write probabilities with 4 decimal places (for example 0.5367), unless the question says something else.
3. Exercises 1 to 4, and Exercise 5 parts (a) and (b), are solved by hand.
4. Exercise 5 part (c) is solved in Python with pgmpy in VS Code, like in the lab.
5. Submit two files on Blackboard: these pages with your answers written in the boxes, scanned as one PDF, and your notebook `assignment_bn.ipynb` with all outputs.
6. This is individual work. You can discuss ideas with classmates, but you must write your own answers.

Exercise 1 — Joint distribution (10 marks)

The Effat shuttle bus is either **on time** or **late**. The weather in Jeddah is either **clear** or **dusty**. The table gives the probability of each pair. For example, the probability that the weather is clear **and** the bus is late is 0.14.

Weather	Shuttle = on time	Shuttle = late
clear	0.56	0.14
dusty	0.12	0.18

1. **(2 marks)** Show that the four numbers in the table make a valid probability distribution.

0.56 + 0.14 + 0.12 + 0.18
= 1.0000

2. **(2 marks)** Find P(dusty), the probability that the weather is dusty. Then find P(late), the probability that the bus is late.

P(dusty) = 0.12 + 0.18 = 0.3000
P(late) = 0.14 + 0.18 = 0.3200

3. **(2 marks)** Find P(late | dusty) and P(late | clear). P(A|B) = P(A and B) / P(B)

P(late | dusty) = P(dusty, late) / P(dusty)
=> 0.18 / 0.30 = 0.6000
P(late | clear) = P(clear, late) / P(clear)
= 0.14 / 0.70 = 0.2000

4. **(2 marks)** Find P(dusty | late). Is it the same as P(late | dusty)? Explain your answer in one sentence.

P(dusty | late) = P(late, dusty) / P(late)
=> 0.18 / 0.32 = 0.5625
=> They're not the same. Both use the same top number (0.18), but one divides by P(late) = 0.32 & the other devides by P(dusty) = 0.30

5. **(2 marks)** Are the weather and the bus independent? Use numbers to explain your answer.

P(late) = 0.3200 0.6000 ≠ 0.3200
P(late | dusty) = 0.6000
=> NO, weather and bus arrival are not independent because P(late | dusty) = 0.6000 is different from P(late) = 0.3200

Exercise 2 — Bayes' rule: a spam filter (10 marks)

An email program wants to find spam emails. It knows these facts:

- 30% of all emails are spam. So P(spam) = 0.3 and P(normal) = 0.7.
- In spam emails, the word "free" appears 60% of the time: P(free | spam) = 0.6.
- In normal emails, the word "free" appears 5% of the time: P(free | normal) = 0.05.

1. **(3 marks)** Find P(free), the probability that any email contains the word "free". Write the name of the rule you use.

P(free) = P(free|spam) P(spam) + P(free|normal) P(normal)
=> (0.6)(0.3) + (0.05)(0.7) = 0.2150

Rule used :- Conditioning

2. **(3 marks)** A new email contains the word "free". Use Bayes' rule to find P(spam | free).

P(spam|free) = P(free|spam) P(spam) / P(free)
= (0.6)(0.3) / (0.2150)
= 0.8372

3. **(3 marks)** The program also checks the word "winner": P(winner | spam) = 0.4 and P(winner | normal) = 0.02. Assume that, once we know if an email is spam or normal, the two words appear independently of each other. A new email contains **both** "free" and "winner". Find P(spam | free, winner).

=> P(free|spam) × P(winner|spam)
= 0.60 × 0.40 = 0.2400
=> P(free|normal) × P(winner|normal)
= 0.05 × 0.02 = 0.0010
=> Spam:- 0.2400 × 0.30 = 0.0720
=> normal:- 0.0010 × 0.70 = 0.0007
P(spam|free,winner) = α(0.0720, 0.0007)

-> α = 1 / (0.0720 + 0.0007) = 13.7552
<0.0720(13.7552), 0.0007(13.7552)>
= <0.9904, 0.0096>
check:- 0.9904 + 0.0096 = 1
P(spam|free,winner) = 0.9904

4. **(1 mark)** Compare your answers to parts 2 and 3. Did the probability of spam go up or down when we added the word "winner"? Explain why in one sentence.

P(spam|free) = 0.8372 P(spam|free,winner) = 0.9904
0.9904 > 0.8372
=> The probability of spam increased from 0.8372 to 0.9904 because the word "winner" is more likely to appear in spam emails (0.40) than in normal emails (0.02).

Exercise 3 — Bayesian network: structure and joint probability (10 marks)

In Jeddah summers, a **heatwave** (H) can cause an **AC overload** (A) in a campus building. A heatwave and an AC overload can both cause a **power cut** (C). A power cut can cause the **lab session to be cancelled** (L). Each variable is yes or no.

The network has 4 arrows: H → A, H → C, A → C, and C → L. The tables give only the probability of "yes". The probability of "no" is 1 minus the probability of "yes".

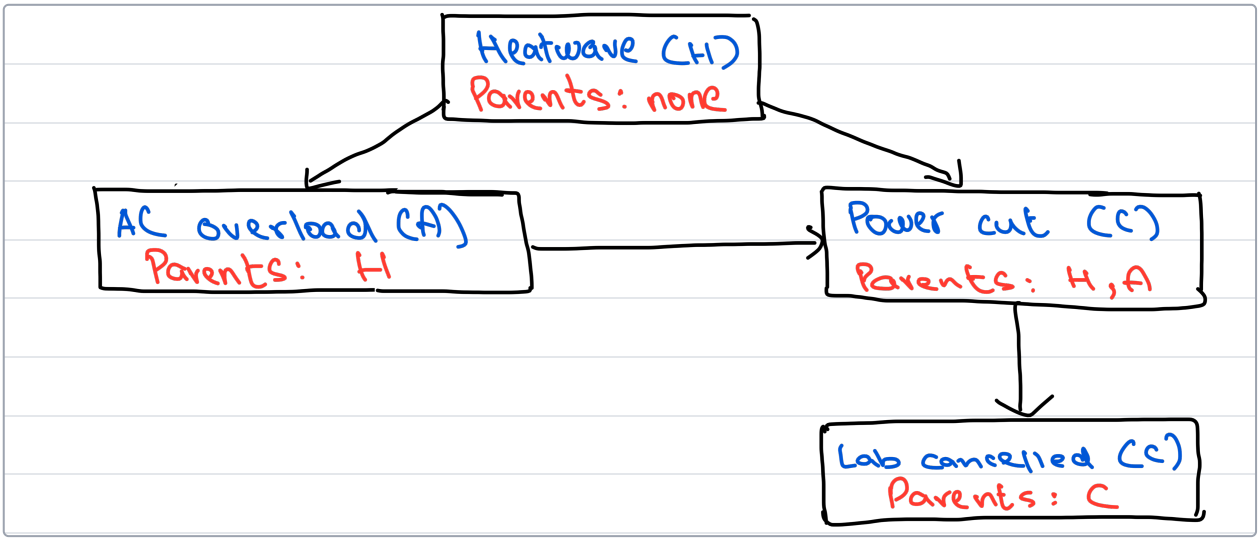
P(H = yes)
0.3

H	P(A = yes | H)
yes	0.6
no	0.1

H	A	P(C = yes | H, A)
yes	yes	0.7
yes	no	0.2
no	yes	0.5
no	no	0.05

C	P(L = yes | C)
yes	0.8
no	0.1

1. **(2 marks)** Draw the network. For each node, write its parents.



2. **(2 marks)** Write P(H, A, C, L) as a product of the probabilities from the network's tables.

$$P(H, A, C, L) = P(H)\,P(A \mid H)\,P(C \mid H, A)\,P(L \mid C)$$

3. **(2 marks)** A full joint table for 4 yes/no variables needs 15 numbers ($2^4 - 1$). How many numbers do the 4 tables of this network need? Count only the "yes" probabilities, and show how you counted.

H:- no parents => $2^0 = 1$
A:- 1 parent => $2^1 = 2$
C:- 2 parents => $2^2 = 4$
L:- 1 parents => $2^1 = 2$

→ Total = 1 + 2 + 4 + 2 = 9*

4. **(2 marks)** Find P(H = yes, A = no, C = yes, L = yes). * use the product from q2.

$P(H = yes) = 0.3$
$P(A = no \mid H = yes) = 1 - 0.6 = 0.4$
$P(C = yes \mid H = yes, A = no) = 0.2$
$P(L = yes \mid C = yes) = 0.8$

$0.3 \times 0.4 \times 0.2 \times 0.8 = 0.0192$

5. **(2 marks)** Find P(H = yes | C = yes). Then give P(H = yes | C = yes, L = yes) without doing a new calculation, and explain why the two answers are the same.

$P(H = yes \mid C = yes) = P(H = yes, C = yes) \mid P(C = yes)$
$\Rightarrow P(H = yes, A = yes, C = yes) + P(H = yes, A = no, C = yes)$
$0.3(0.6)(0.7) + 0.3(0.1)(0.2) = 0.1500$
$\Rightarrow P(H = no, C = yes) = 0.7(0.1)(0.5) + 0.7(0.9)(0.05) = 0.0665$
$P(H \mid C = yes) = \alpha \langle 0.1500, 0.0665 \rangle \Rightarrow \alpha = \frac{1}{0.1500 + 0.0665} = 4.6154$
$= \langle 0.6923, 0.3077 \rangle$
$P(H = yes \mid C = yes) = 0.6923$

Exercise 4 — Inference by enumeration, challenge question (10 marks)

This is the **challenge question** of the assignment. Show every step.

Use the network and tables from Exercise 3. Today the lab session **was cancelled** (L = yes).

1. **(4 marks)** Use inference by enumeration to find P(H = yes | L = yes), the probability that there was a heatwave. Write the query variable, the evidence and the hidden variables, then show all sums and the normalization.

$P(H=yes, L=yes) = \sum_A \sum_C P(H=yes, A, C, L=yes)$

$\Rightarrow H = yes$

A	C	$P(H)P(A\mid H)P(C\mid H,A)P(L=y\mid C)$	
yes	yes	$0.3 \times 0.6 \times 0.7 \times 0.8$	$= 0.1008$
yes	no	$0.3 \times 0.6 \times 0.3 \times 0.1$	$= 0.0054$
no	yes	$0.3 \times 0.4 \times 0.2 \times 0.8$	$= 0.0192$
no	no	$0.3 \times 0.4 \times 0.8 \times 0.1$	$= 0.0096$
			$= 0.1350$

$\Rightarrow H = no$

A	C	Product	
yes	yes	$0.7 \times 0.1 \times 0.5 \times 0.8$	$= 0.0280$
yes	no	$0.7 \times 0.1 \times 0.5 \times 0.1$	$= 0.0035$
no	yes	$0.7 \times 0.9 \times 0.05 \times 0.8$	$= 0.0252$
no	no	$0.7 \times 0.9 \times 0.95 \times 0.1$	$= 0.05985$
			$= 0.11655$

$\Rightarrow$ normalization

$P(L=yes) = 0.1350 + 0.11655 = 0.25155 \Rightarrow P(H=yes \mid L=yes) = \frac{0.1350}{0.25155} = 0.5367$

2. **(3 marks)** Now find P(A = yes | L = yes), the probability that there was an AC overload. Which variables are hidden this time?

$P(A=yes \mid L=yes)$

For $H=yes, A=yes$

$0.3(0.6)[(0.7)(0.8)+(0.3)(0.1)]$

$= 0.18(0.56+0.03) = 0.1062$

For $H=no, A=yes$

$0.7(0.1)[(0.5)(0.8)+(0.5)(0.1)] = 0.0315$

$+ \rightarrow 0.1377$

$P(A=no \mid L=yes)$

For $H=yes \rightarrow 0.3(0.4)[(0.2)(0.8)+(0.8)(0.1)] = 0.0288$

For $H=no \rightarrow 0.7(0.9)[(0.05)(0.8)+(0.95)(0.1)] = 0.08505$

$+ \rightarrow 0.11385$

$\rightarrow$ normalize

$P(A \mid L=yes) = \alpha\langle 0.1377, 0.11385\rangle \quad \alpha = \frac{1}{0.1377+0.11385} = 3.9754$

$\langle 0.1377(3.9754), 0.11385(3.9754)\rangle$

$= 0.5474, 0.4526$

$\Rightarrow P(A=yes \mid L=yes) = 0.5474$

→ The probability of a heatwave decreased from 0.5367 to 0.2530 after observing that there was no AC overload. This is because a heatwave increases the likelihood of an AC overload, so observing no overload makes a heatwave less likely.

3. **(3 marks)** A technician checks the building and reports that there was **no** AC overload (A = no). The lab is still cancelled (L = yes). Find P(H = yes | L = yes, A = no). Compare it with your answer to part 1: did the probability of a heatwave go up or down? Use the network to explain why in two or three sentences.

$P(H=yes \mid L=yes, A=no) = \alpha \sum_C P(H)P(A=no \mid H)P(C \mid H, A=no)P(L=yes \mid C)$

$\Rightarrow H = yes$

C	$P(H=yes) \cdot P(A=no \mid H=yes) \cdot P(C \mid H=yes, A=no) \cdot P(L=yes \mid C)$		
yes	$0.3 \times 0.4 \times 0.2 \times 0.8$	$= 0.0192$	$+$ 0.0288
no	$0.3 \times 0.4 \times 0.8 \times 0.1$	$= 0.0096$	

$\Rightarrow H = no$

C	$P(H=no) \cdot P(A=no \mid H=no) \cdot P(C \mid H=no, A=no) \cdot P(L=yes \mid C)$		
yes	$0.7 \times 0.9 \times 0.05 \times 0.8$	$= 0.0252$	$+$ 0.08505
no	$0.7 \times 0.9 \times 0.95 \times 0.1$	$= 0.05985$	

normalize

$P(H \mid L=yes, A=no) = \alpha \langle 0.0288, 0.08505 \rangle = 0.0288(8.7835), 0.08505(8.7835)$

$\alpha = \frac{1}{0.0288 + 0.08505} = 8.7835 = \langle 0.2530, 0.7470 \rangle$

$\Rightarrow P(H \mid L=yes, A=no) = 0.2530$

Exercise 5 — Approximate inference by sampling (10 marks)

Now we find the same probability, P(H = yes | L = yes), by **sampling**. Sampling gives an estimate, not the exact answer.

(a) Rejection sampling (3 marks). Reminder: in rejection sampling, we keep only the samples that agree with the evidence (here L = yes), and we throw away the others. Below are 12 samples from the network of Exercise 3.

#	H	A	C	L
1	no	no	no	no
2	yes	yes	yes	yes
3	no	no	no	no
4	no	yes	yes	yes
5	yes	yes	no	no
6	no	no	no	yes
7	yes	no	yes	yes
8	no	no	no	no
9	no	no	no	no
10	yes	yes	yes	no
11	no	no	no	no
12	no	no	yes	yes

1. Write the numbers of the samples we keep and the samples we throw away. Then estimate P(H = yes | L = yes).

* keep samples L = yes
throw = L = no
=> kept → 2, 4, 6, 7, 12
=> thrown away → 1, 3, 5, 8, 9, 10, 11
H = yes among kept samples = 2
P(H = yes | L = yes) = 2/5 = 0.4000

2. What fraction of the 12 samples did we throw away? Why is this a problem for rejection sampling?

we threw away 7 samples out of 12
7/12 = 0.5833 → 58.33%

=> We threw away 58.33% of the samples. This is a problem because many are wasted, leaving fewer samples to estimate the probability accurately

(b) Likelihood weighting (3 marks). Reminder: in likelihood weighting, we do not sample the evidence. We fix it (here L = yes), we sample the other variables, and we give each sample a **weight**. The weight is the probability of the evidence given the sample's values: here, P(L = yes | C). Below are 6 samples.

#	H	A	C	L
1	no	no	no	yes (fixed)
2	yes	yes	yes	yes (fixed)
3	no	no	no	yes (fixed)
4	yes	no	no	yes (fixed)
5	no	yes	yes	yes (fixed)
6	yes	yes	yes	yes (fixed)

1. Write the weight of each sample.

P(L = yes | C = yes) = 0.8 P(L = yes | C = no) = 0.1

#	C	weight
1	no	0.1000
2	yes	0.8000
3	no	0.1000
4	no	0.1000
5	yes	0.8000
6	yes	0.8000

2. Estimate P(H = yes | L = yes) using the weights: add the weights of the samples where H = yes, then divide by the sum of all weights.

$$P(H=\text{yes} \mid L=\text{yes}) = \frac{\text{weight where } H=\text{yes}}{\text{sum of all weights}} = \frac{0.8+0.1+0.8}{0.1+0.8+0.1+0.1+0.8+0.8}$$

$$= \frac{1.7}{2.7} = 0.6296$$

3. Compare your estimates from (a) and (b) with the exact answer from Exercise 4. Why are they different? How can we make them closer to the exact answer?

exact ans: 0.5367
Rejection sampling: 0.4000
Likelihood weighing: 0.6296

=> The estimates are different from the exact answer because sampling uses a limited no. of samples, which can cause random variations.
We can make the estimates closer to the exact answer by increasing the number of samples.

(c) Python with pgmpy (4 marks). Answer this part in your notebook, not on paper. In VS Code, create `assignment_bn.ipynb`.

1. Build the network of Exercise 3 with `DiscreteBayesianNetwork` and `TabularCPD`, like in the lab. Name the nodes `heatwave`, `overload`, `powercut` and `cancel`, and use the values `yes` and `no`. Print `model.check_model()`. It must print `True`.
2. Use `VariableElimination` to find P(H | L = yes). Your answer must be the same as in Exercise 4.
3. Use rejection sampling with 10,000 samples to estimate P(H = yes | L = yes). Hint: import `BayesianModelSampling` from `pgmpy.sampling` and `State` from `pgmpy.factors.discrete`. Then use `BayesianModelSampling(model).rejection_sample(evidence=[State("cancel", "yes")], size=10000)`. This gives a table of samples. Count the fraction of rows where `heatwave` is `yes`.
4. In a Markdown cell, write two sentences that compare the exact answer and the sampling estimate.